



Image Segmentation: Segment Anything Model (SAM)

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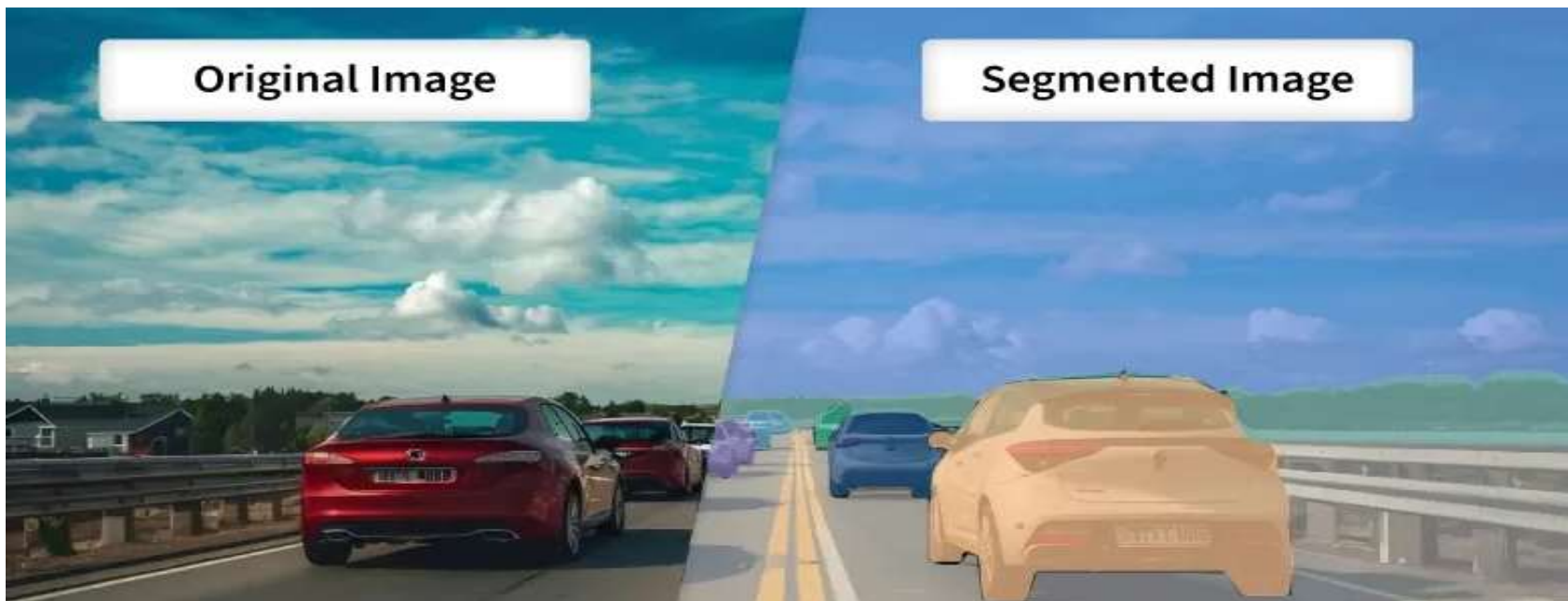
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Image Segmentation –What?

- Image segmentation is a technique that partitions a digital image into multiple segments or regions, typically by assigning a class label to every individual pixel.
- This pixel-level classification allows machines to identify object boundaries, separate foregrounds from backgrounds, and understand the layout of an image.





Some Image Segmentation Applications?

- **Self-driving cars:** It help the driving instruction algorithm to better assess the surrounding before generating next instruction.
- Image segmentation is used for easy distinctions between various objects. Be it traffic signals, signboards, humans, and cars.





Industrial inspection



- A company has to bear the responsibility of defected devices. If a camera backed with an Image Segmentation model keeps scanning for defects produced in the final product, a lot of money and time can be saved in fixing a defective device. Example:
Circuit Board Defect Detection:

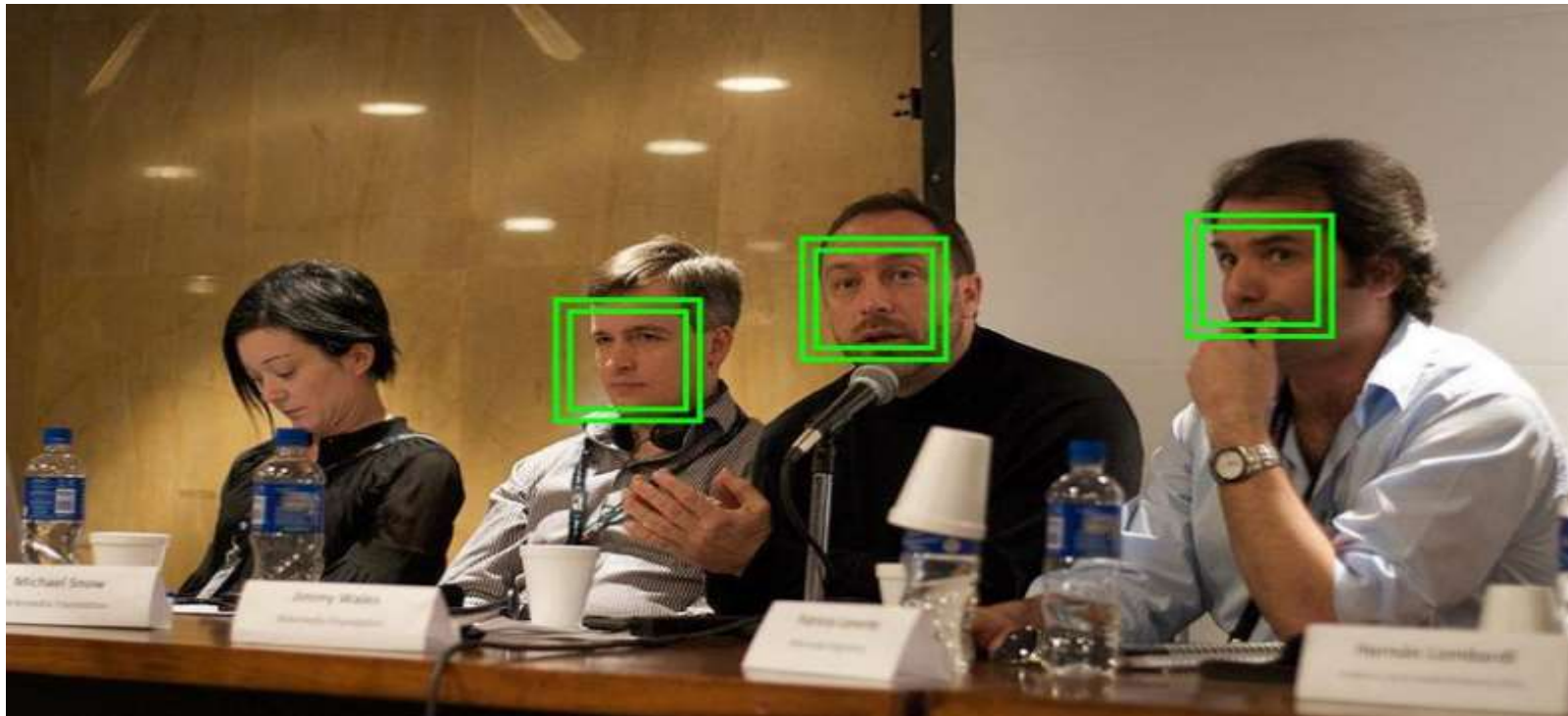




Face Detection



- The majority of cameras in phones support portrait mode. Portrait mode is technically an outcome of Image Segmentation. Apart from this, security surveillance will be much more effective when the faces are distinguishable from noisy objects.

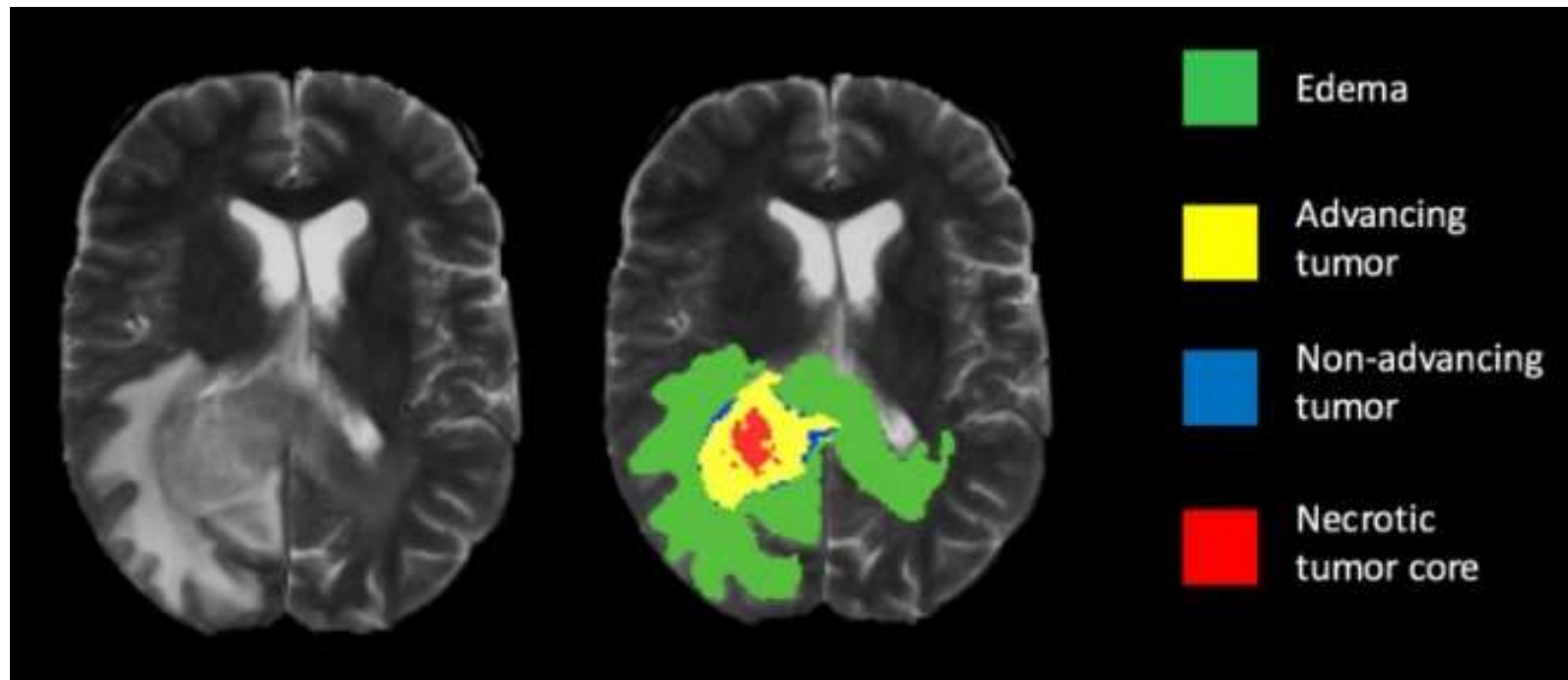




Medical Imaging



- Image segmentation can be used to extract clinically relevant information from medical reports. For example, image segmentation can be used to segment tumors.





Types of Image Segmentations

- Image Segmentation is broadly classified into three types:

1. **Semantic Segmentation:** It is segmenting the image pixels into their respective classes. For example, the pixels related to the car are colored blue, related to humans are colored brown, related to trees are colored green, etc. Multiple objects of the same class are considered as a single entity.



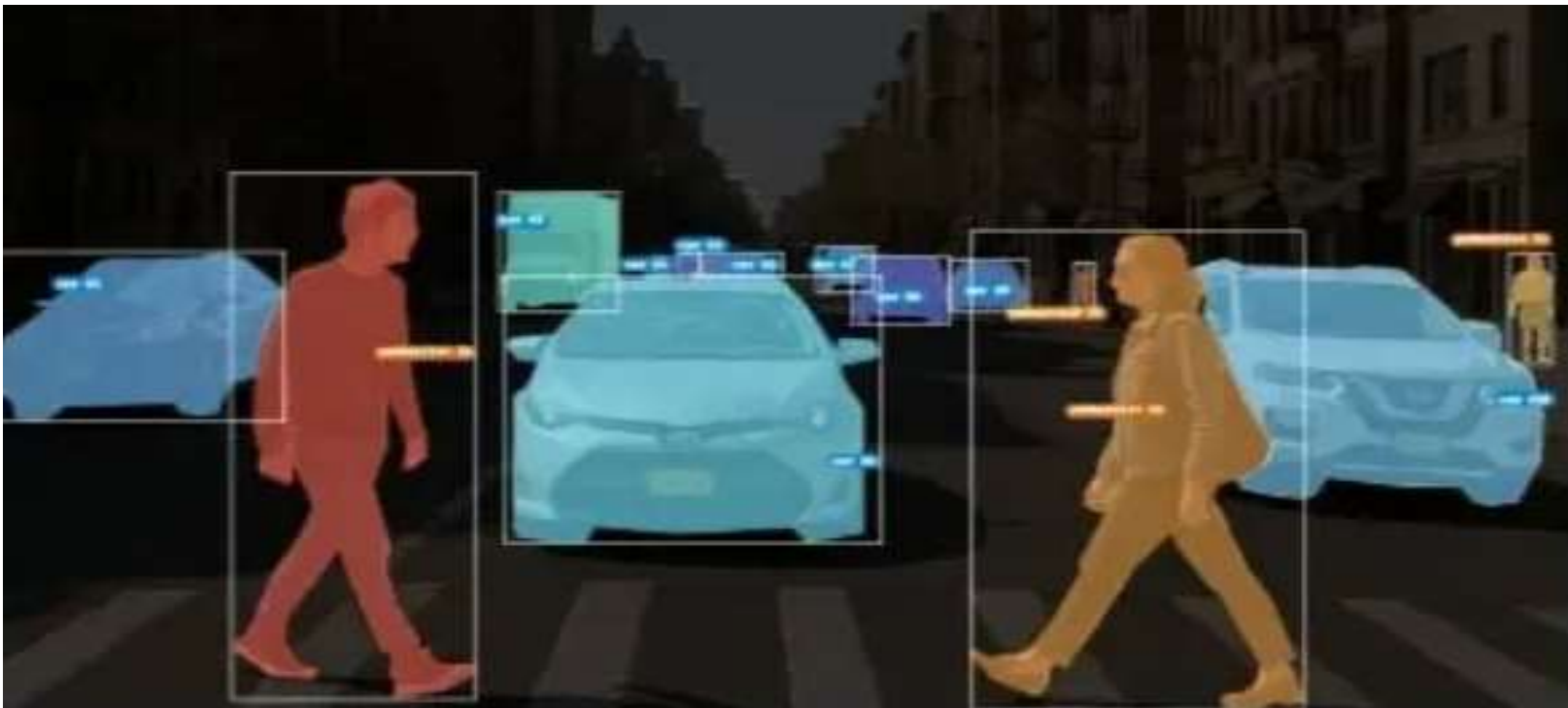


Types of ...



2. **Instance Segmentation**: It usually comes into picture when dealing with multiple objects.

Multiple objects of the same class are treated as distinct entities and hence represented with different colors.





Types ...

3. Panoptic Segmentation: Combines semantic and instance segmentation. It assigns a class label to every single pixel in the image while still detecting distinct, countable objects





Segment Anything Model (SAM)

- **SAM** is a Meta AI, the parent company of facebook, product.
- **SAM** is first developed about three years ago.
- Up to now, three generation of SAM are introduced, **SAM1**, **SAM2**, **SAM3**.
- **SAM** is a *promptable, foundation model* for image segmentation.



Overview (continued)

- **SAM** can segment *any* object in an image using simple prompts (points, boxes, or masks) and generalizes to unseen objects without retraining.
- It is built from three main components:
 - a **Vision Transformer image encoder**
 - a **prompt encoder**,
 - and a **mask decoder**.
- *Note: some of these advanced concepts are not covered in the course. But, **SAM** as a exceptionally valuable and universally available tool, is discussed here.*
 - *Experimenting **SAM** as a tool, is the main objective here.*



What SAM Is

- **SAM** is designed to make segmentation both
 - **Universal**
 - **Interactive**
- **Open-source**: Code, pre-trained weights, and demos are publicly available in Github:
 - <https://github.com/facebookresearch/segment-anything>
- This makes **SAM** ideal because academics can experiment immediately without building datasets or training models.



Why It Matters

- **Promptable segmentation:** Users guide segmentation with clicks, bounding boxes, texts, or previous masks.
- **Zero-shot generalization:** SAM can segment objects it has never seen before.
- **Massive training dataset:** Trained on **11 million images** and **1.1 billion masks** (SA-1B dataset).
- **High-quality masks:** Produces clean, pixel-accurate boundaries even in complex scenes.



SAM Architecture (Simple)



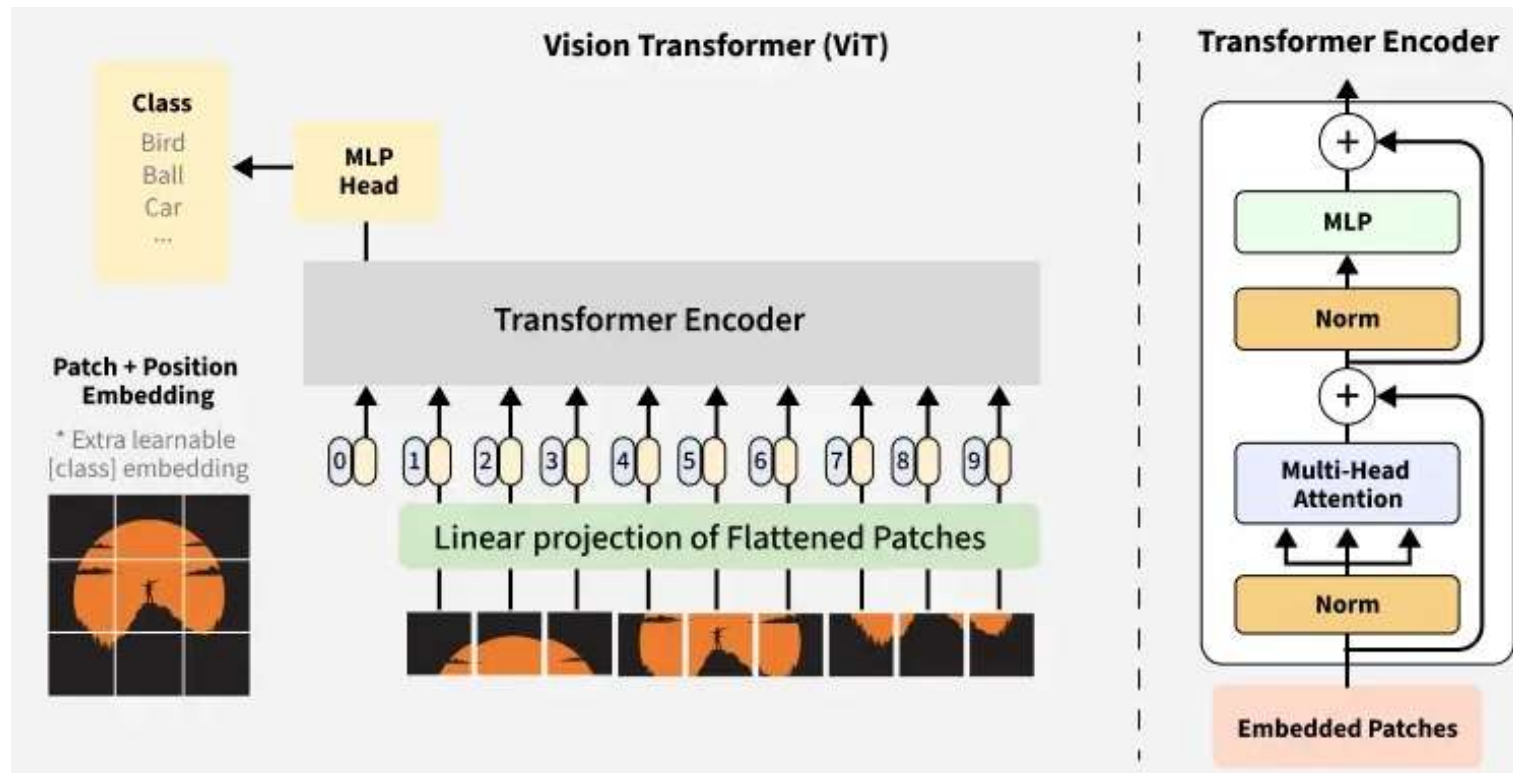
A. Vision Transformer (ViT) Image Encoder

- Splits the image into patches and processes them like tokens.
- Produces a dense embedding representing the whole image.
- Vision Transformer (ViT) is a deep learning architecture that applies the **Transformer** model to images.
- Instead of relying on convolutions, ViTs use **Self-Attention** to capture relationships across all image patches, enabling a global understanding of the image.
- This approach has achieved state-of-the-art results in various computer vision tasks.
- Uses **Self-Attention** to model global dependencies between image patches.



ViT model

- ViT treats an image as a sequence of fixed-size patches and applies self-attention across them.
- This allows the model to capture long range dependencies between different parts of an image without relying on convolution operations.



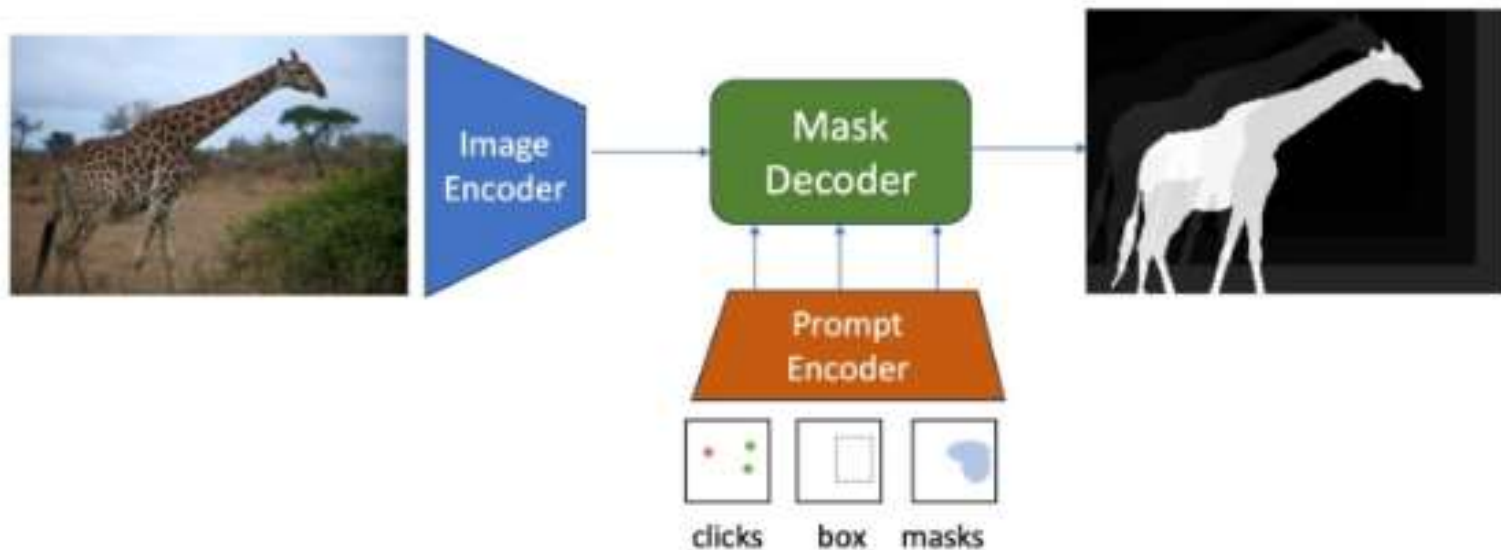


SAM Architecture (Continued ...)



B. Prompt Encoder

- The **Prompt Encoder** in SAM translates user inputs (**Clicks or Points, bounding boxes, or text**) into mathematical vector embedding.
- It bridges the gap between human intent and the visual data processed by the model.





SAM Architecture (Continued ...)

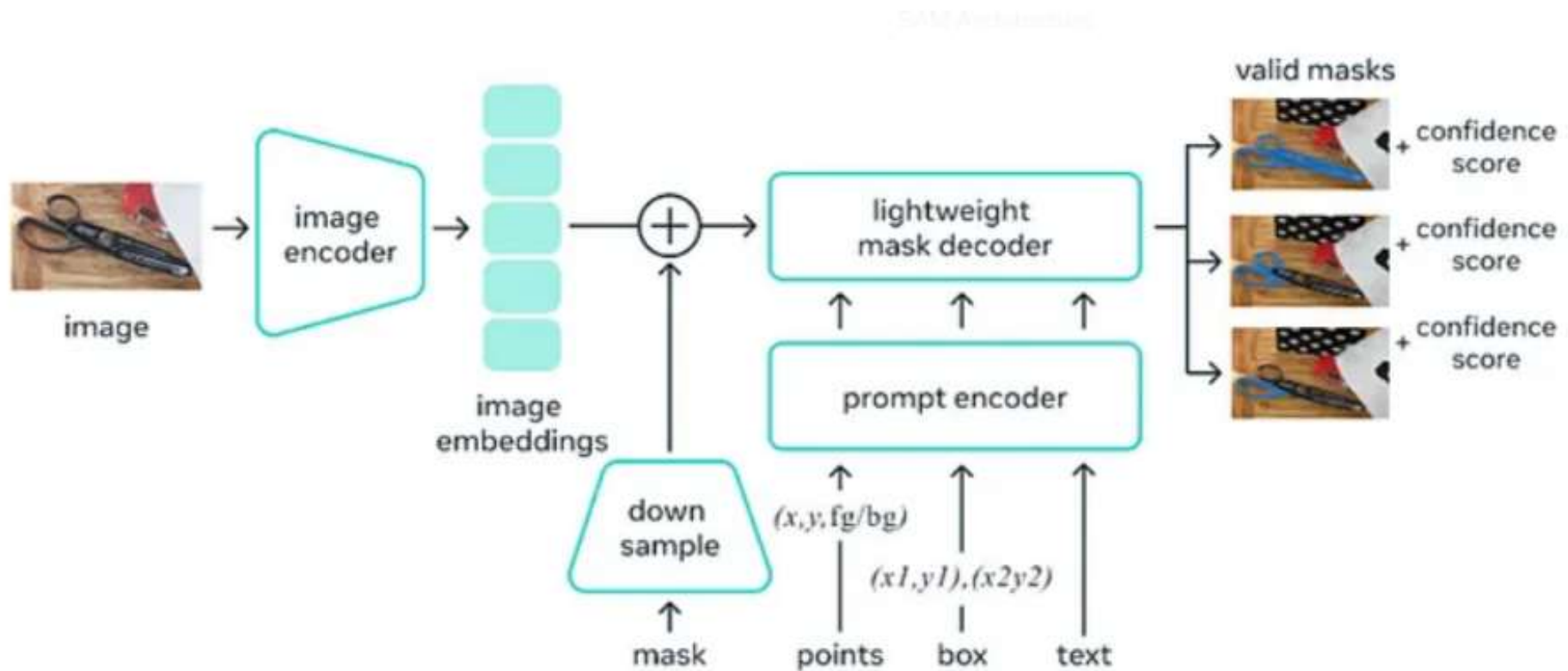
C. Mask Decoder

- A highly efficient transformer-based module that acts as the bridge between visual information and user prompts (Lightweight network that combines image + prompt embeddings)
- Its primary job is to take pre-computed image embeddings and merge them with prompt tokens (points, boxes, text, or rough masks) to output high-quality, pixel-level segmentation masks in real time.
- This modular design makes SAM easy to explain and visualize in class.



SAM overall Architecture Scheme

- **Segment Anything Model** contains 3 components: **image encoder**, **prompt encoder**, and **mask decoder**

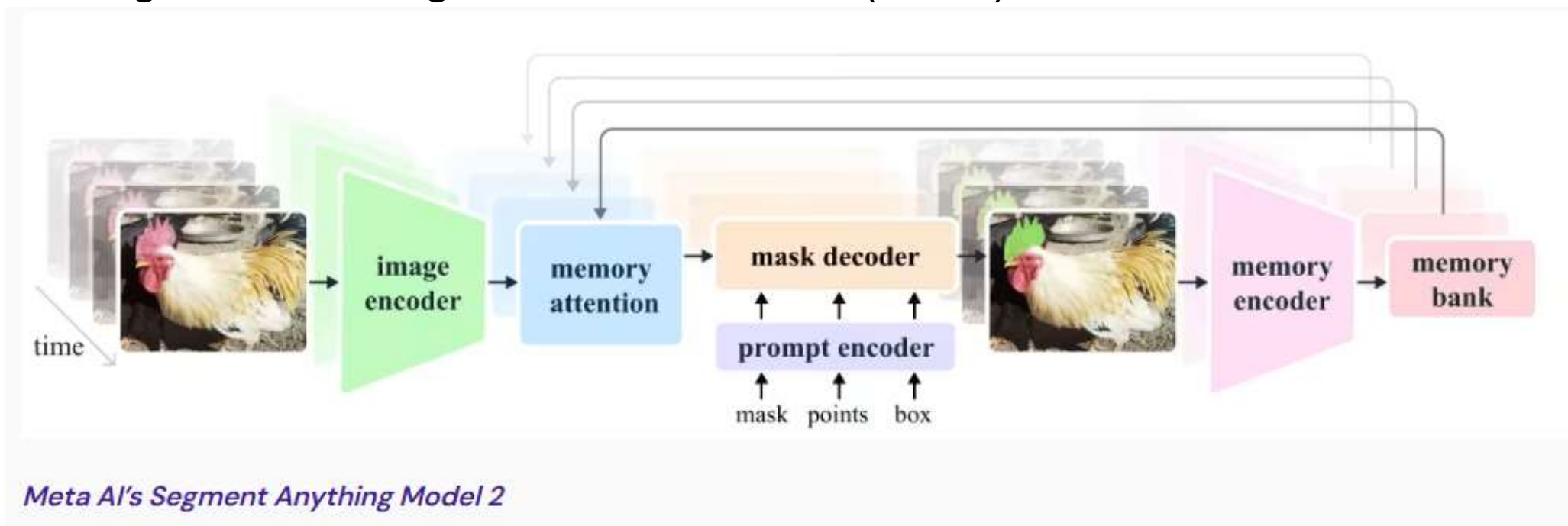




SAM 2 — What's New



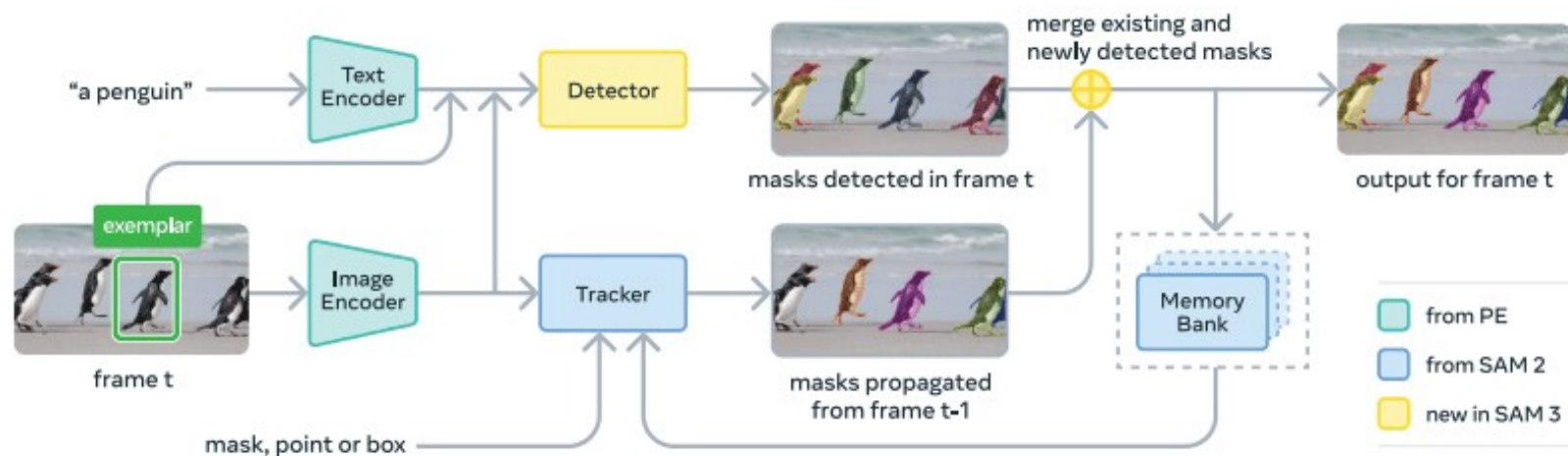
- Meta released **SAM 2**, extending segmentation to **videos**:
- Treats images as single-frame videos.
- Uses a **transformer with streaming memory** for real-time video segmentation.
- Built with a **model-in-the-loop data engine** to collect the largest video segmentation dataset (SA-V).





SAM 3 – What is New?

- SAM 3 shifts segmentation from a manual, geometric task to an **open-vocabulary, concept-based vision foundation model**.
- It can detect, segment, and track all instances of a concept using natural language or image exemplars, eliminating the need for clicks and boxes

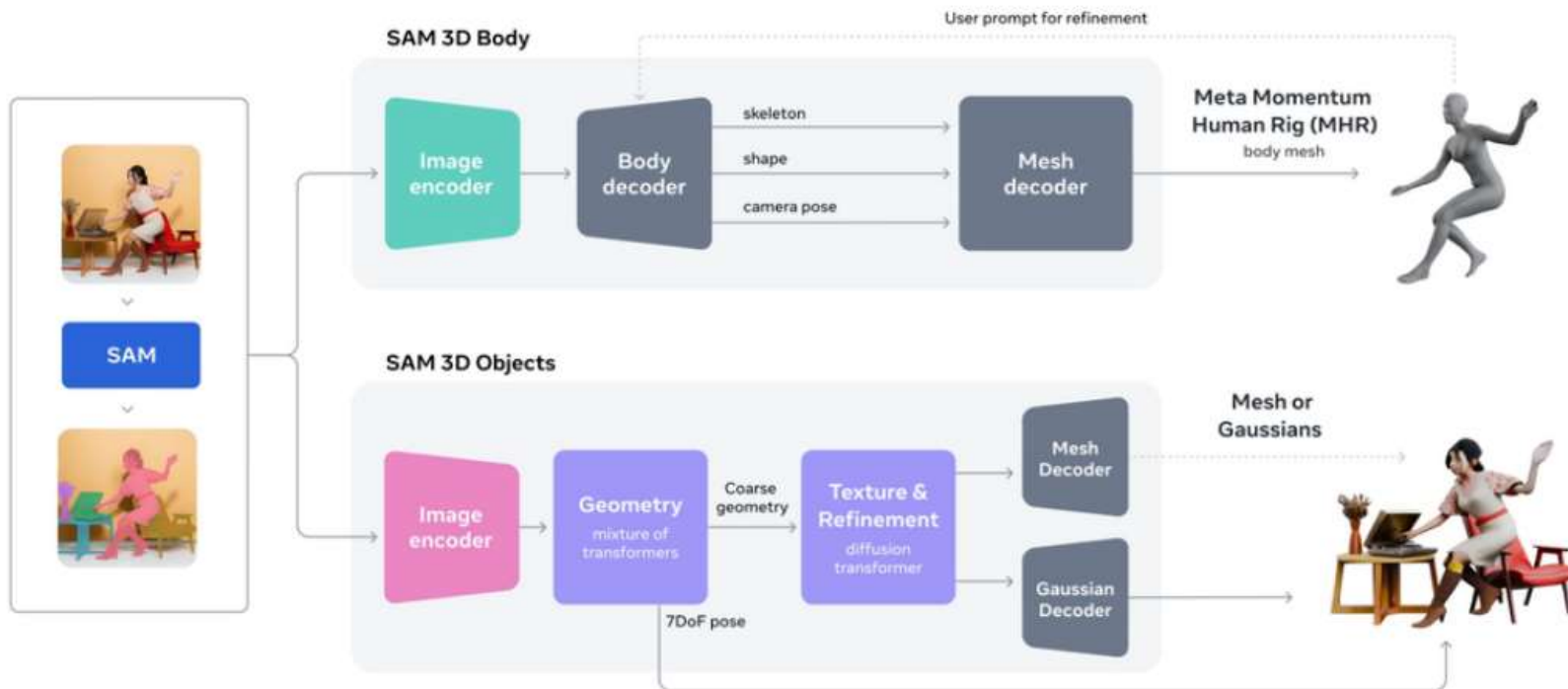




SAM 3D Companion

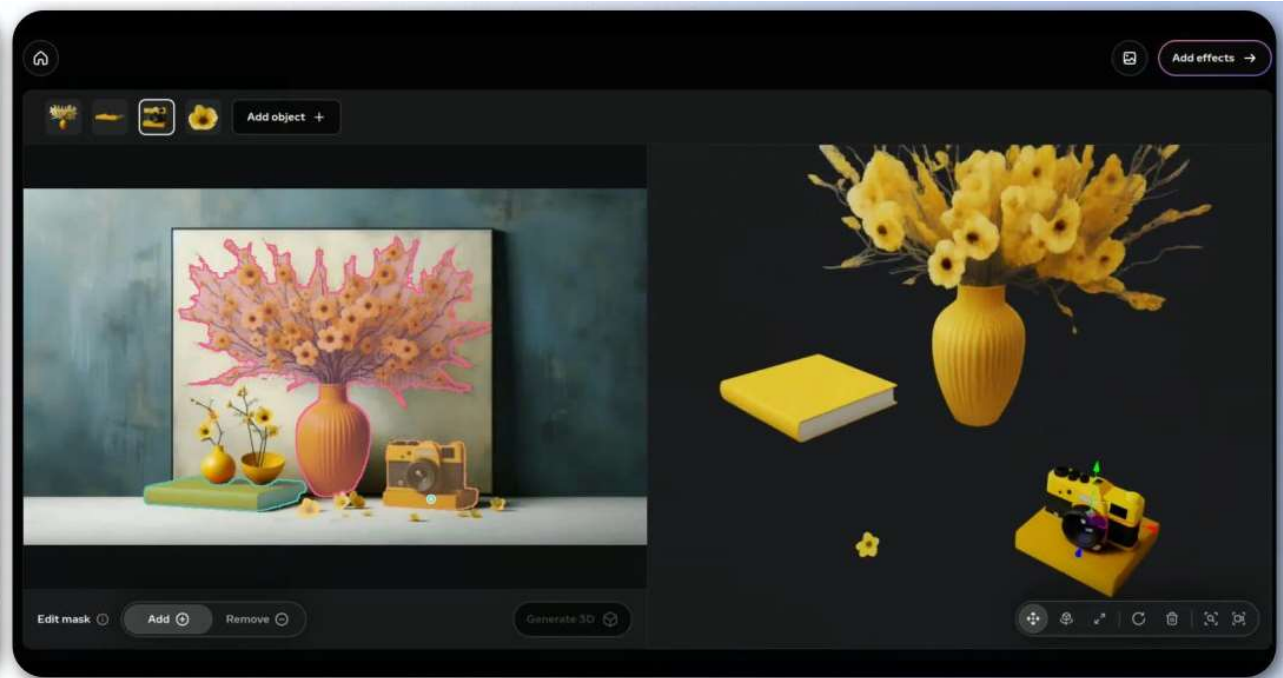


- **SAM 3D Companion Models:** Alongside SAM 3, Meta released SAM 3D, a suite of models capable of reconstructing 3D objects and human bodies with their textures and geometry from a single 2D image.

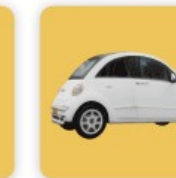




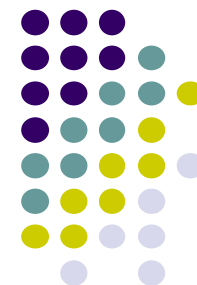
INPUT IMAGE



SINGLE IMAGE 3D MODELS RECONSTRUCTION



SAM 3D reconstructs diverse objects and scenes from a single image with accurate geometry and texture



Final Project: A project on SAM

عنوان پروژه:

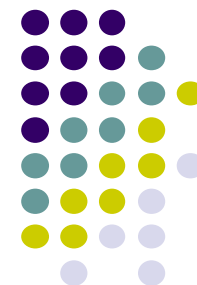
ردیابی ویدئویی با واژگان باز (open vocabulary)

هدف:

پیاده‌سازی قابلیت‌های مبتنی بر متن در SAM 3 / 3.1 برای تفکیک یک هدف متحرک در یک کلیپ ویدیویی و محاسبه ماتریس حرکت آن.

آدرس SAM3:

<https://github.com/facebookresearch/sam3>

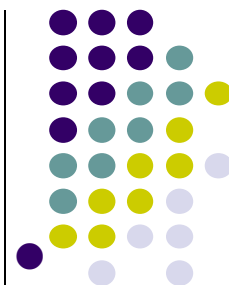


برنامه نویسی پایتون پروژه:

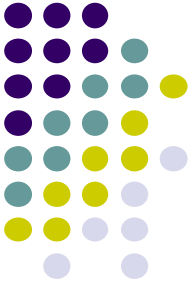
- یک ویدیوی کوتاه ۵ ثانیه‌ای (MP4) شامل یک هدف متحرک تهیه کنید.
 - با استفاده از موتور دستوردهی متنی موجود در مخزن GitHub مربوط به SAM 3، آن‌ها باید هدف را ردیابی کرده، مرکز (centroid) کادر محصورکننده آن را در تمام فریم‌ها استخراج کنند و مسیر حرکت آن را ترسیم نمایند.
 - راه‌اندازی پروژه در Google Colab :
 - پیاده سازی pipe line پردازش ویدیویی مبتنی بر متن در SAM 3 به شرح زیر:
 - ❖ مقداردهی اولیه مدل پیش‌بینی ویدیویی ارائه یک دستور متنی (مثلاً "توپ فوتبال")
 - ❖ برای تولید ماسک اولیه در فریم گسترش وضعیت ردیابی در تمام فریم‌های ویدیو
 - ❖ برای هر فریم، محاسبه مرکز ماسک:
- $$cX = \text{int}(M["m10"] / M["m00"])$$
- $$cY = \text{int}(M["m01"] / M["m00"])$$
- خروجی گرفتن از ویدیو پردازش شده همراه با کادر محصورکننده و مسیر حرکت هدف روی آن



آنچه شما تحویل می دهید:



- گزارش کتبی شامل کد و تجزیه و تحلیل از روند انجام و نتایج بدست آمده،
- ارائه یک مخزن GitHub شامل نوتبوک Colab ،
- یک فایل README.md که شامل دستور العمل پیاده سازی و اجرای کدها باشد،
- یک GIF یا کلیپ MP4 رندر شده که نشان می دهد هدف به صورت تمیز ماسک شده و در حال حرکت است،
- یک نمودار پراکندگی دوبعدی که مختصات پیکسلی $[X, Y]$ شیء را در طول زمان نشان می دهد و مسیر دقیق حرکت آن در کلیپ ویدیویی را ترسیم می کند.
- **آخرین فرصت برای تحویل این پروژه، همزمان با امتحان پایان ترم است.**
- تمام موارد (بجز مخزن) در قالب یک CD تحویل داده شوند. همراه با CD، یک برگ A4 شناسه شامل مشخصات پروژه، مشخصات دانشجو، آدرس مخزن و چکیده گزارش پروژه تحویل داده شود. تحویل پروژه در شبکه های اجتماعی، قابل قبول نیست.
- **نکته مهم:** در صورت اثبات هر گونه تقلب شامل برون سپاری پروژه، استفاده نامعقول از ابزارهای AI، کپی از دیگران و برای شما نمره صفر در نظر گرفته خواهد شد. تشخیص تقلب صرفا بعهدہ اینجانب بوده و تصمیم بنده نهایی است.



Thank you

for your attention